

Lesson 11: Power and energy

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.4.1-2

Why this lesson is here

A kettle and a lamp use the same mains supply but transfer energy at different rates. Their power ratings tell you how fast.

Learning objective

Choose the right equation, convert units and solve multi-step appliance problems.

Core teaching point

Power is energy transferred per second. $P = VI$ and $P = I^2R$; $E = Pt$ and $E = QV$. Convert kW to W and minutes/hours to seconds for joules.

Equipment and preparation

Appliance rating labels, calculators and equation toolkit.

Lesson sequence

Time	Phase	Teacher move and evidence
0-6	Retrieval	Quick equations and unit conversion.
6-18	Model	Power from VI and I^2R ; show square-current effect.
18-31	Worked chain	Kettle current then energy with units.
31-43	Independent	Tasks 1-4, then six-mark question.
43-50	Check	Peer mark methods and conversions.

Workbook task guidance

Task 1: Choose an equation for each pair of known quantities: V and I ; I and R ; P and t ; Q and V .

Expected: $P = VI$; $P = I^2R$; $E = Pt$; $E = QV$.

Task 2: A 230 V heater draws 4.0 A. Calculate its power and energy in 5.0 min.

Expected: $P = 920 \text{ W}$; $t = 300 \text{ s}$; $E = 276000 \text{ J}$.

Task 3: A 3.0 A current passes a 20 ohm resistor. Calculate dissipated power; explain why the current is squared.

Expected: $P = 3.0^2 \times 20 = 180 \text{ W}$; doubling I quadruples heating at fixed R .

Task 4: A device transfers 240 J while 40 C flows. Calculate its p.d.

Expected: $V = E/Q = 240/40 = 6.0 \text{ V}$.

Worked example

2.0 kW kettle for 3.0 min: $P = 2000 \text{ W}$, $t = 180 \text{ s}$, $E = Pt = 360000 \text{ J}$.

AQA-style checkpoint

[6 marks] A 2.2 kW kettle runs at 230 V for 150 s. Calculate current and energy transferred in joules. A second kettle draws twice the current through the same resistance: compare its heating power.

Mark guidance: $P = 2200 \text{ W}$ (1); $I = P/V = 2200/230 = 9.6 \text{ A}$ (2); $E = Pt = 2200 \times 150 = 330000 \text{ J}$ (2); $P = I^2R$ so doubling I gives four times the heating power (1).

Anticipated misconception

$P = I^2R$, not IR ; $E = Pt$ needs W and s for J .

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.